Information Entropy Estimation Based on Point-Set Topology for Hyperspectral Anomaly Detection

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***Abstract*— As one of the most active research hotspots in hyperspectral remote sensing, anomaly detection is widely used because it takes effect without any priori information about the target or the background. Most of the traditional model-driven methods fail to reveal features of data with diversity due to fixed analytical modes. A variety of data-driven methods encounter difficulties in practical applications due to their costly computational complexity. In this article, an innovative combination of point-set topology and information entropy theories is utilized to analyze the mathematical–statistical properties of hyperspectral images (HSIs), thus eliminating the limitations caused by the data-model discrepancy. Specifically, the original HSI data are mapped into topological spaces in a specific form to enable ordered arrangements, in preparation for revealing data features. In particular, information entropy estimation is introduced for the first time in the adoption of point-set topology to adequately unravel the data arrangements in topological spaces, whereby the land cover information is efficiently extracted for detection. Accordingly, an interesting approach of information entropy estimation based on point-set topology (IEEPST) is proposed to resolve anomaly detection from a brand new perspective, pursuing prominent detection accuracy while ensuring computational efficiency. The experimental results on benchmark HSI datasets demonstrate that IEEPST achieves detection performance with high probabilities of detection (PD) and low false alarm rates (FARs) at an inexpensive computational cost. The proposed IEEPST is highly competitive with other sophisticated and state-of-the-art methods.**

***Index Terms*— Anomaly detection, data mining, hyperspectral image, information entropy, topology.**

Manuscript received 31 August 2023; revised 11 January 2024, 10 April 2024, and 21 May 2024; accepted 2 July 2024. Date of publication 8 July 2024; date of current version 22 July 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 42325104 and Grant 62161160336. *(Corresponding author: Lianru Gao.)*

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Digital Object Identifier 10.1109/TGRS.2024.3424465

I. Introduction

HYPERSPECTRAL remote sensing integrates spectral information reflecting material with spatial information reflecting distribution for various classes of land covers [1], [2], [3], [4]. Each pixel in the acquired image corresponds to an approximately continuous spectrum, enabling the acquisition of detailed information about land covers [5], [6], [7], [8]. Target detection in hyperspectral images (HSIs) is capable of locating objects of interest within a specific scene based on spectral characteristics [9], [10]. Hyperspectral target detection can be categorized into supervised matching detection and unsupervised anomaly detection depending on the availability of the priori spectral information about the target [11], [12], [13]. Anomaly detection is of great value in mineral exploration, atmospheric composition detection, vegetation growth monitoring, battlefield target reconnaissance, mine detection, target counter-camouflage, and other applications covering both military and civilian due to the ease with which the conditions of validity can be met [14], [15], [16], [17]. Given the irreplaceable role of hyperspectral anomaly detection in a variety of practical applications, the development of detection methods suitable for HSIs is essential.

Signal processing theories have been extensively applied to the processing and analysis of HSIs since the 1990s and a wide range of algorithms have been developed to meet specific application requirements. The most classic and widely used traditional methods are derived based on probabilistic statistical models and subspace projections, including Reed–Xiaoli detector (RXD) [18], low probability target detector (LPTD), uniform target detector (UTD) [14], unsupervised vector quantization (UVQ), and unsupervised target generation process (UTGP), [19]. In [20], an interesting detector without requiring any supervised information was designed for hyperspectral anomaly detection, according to the two-step version of generalized likelihood ratio test. This unsupervised detector exhibits better detection performance than other unsupervised approaches. In addition, the Bayesian Gaussian mixture model (GMM) split-based (BGMMS) anomaly detector [21], which is derived from a more sophisticated model [22], significantly enhances the accuracy of anomaly detection. Furthermore, a series of algorithms derived from RXD have exhibited advantages of efficiency and robustness in enormous applications, such as global RXD (Global-RXD) [18], local RXD

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(Local-RXD) [18], segmented RXD (Segmented-RXD) [23], and iterative RXD (Iterative-RXD) [24]. Despite the high interpretability and low computational costs, the performance of traditional methods is severely limited by the discrepancy between specific model assumptions and HSIs being processed.

With the variability of imaging scenes and the improvement in the quality of image data available nowadays, the introduction of machine learning brings fresh perspectives to the solution of hyperspectral anomaly detection. Several categories of research hotspots have yielded a wealth of high-performance detectors [25]. Among them, representative methods of the representation category [26], such as collaborative representation-based detector (CRD) and low-rank and sparse representation-based detector (LRASR), represent pixels in HSIs by constructing multiple forms of dictionaries to achieve an accurate anomaly-background separation [27], [28], [29]. The design of clustering-based detection methods is based on the fact that anomalies occupy small areas with fewer pixels, while the background comprises multiple clusters. Taking advantage of this, such methods aim to better learn the background and improve the detection performance. Representative examples include density peak (DP) clustering-based anomaly detector and clustering-based background learning (CBL) anomaly detector [30], [31]. Robust nonlinear anomaly detection (RNAD) and kernel RXD (Kernel-RXD) adopt various kernel mappings to enhance the capability of detectors for processing nonlinear data [32], [33]. In particular, cluster Kernel-RXD (CKRX) combines kernel methods and clustering, where the key idea is to group background pixels into clusters and then apply a fast eigendecomposition algorithm to generate the anomaly detection index [34]. In addition, there are methods for estimating the background in HSIs through kernel density estimation, and they are fully nonparametric, facilitating the convenient localization of anomalies in practical applications [35], [36]. Algorithms, such as subspace selection-based discriminative forest (SSDF) and kernel isolation forest-based hyperspectral anomaly detector (KIFD) developed based on graph theory, utilize the connectivity between nodes in a particular graph to achieve the anomaly-background separation and to design the detection output [37], [38]. Topology-based methods such as tree topology-based anomaly detector (TTAD) and chessboard topology-based anomaly detector (CTAD) map the HSI data from the original space into the topological space in a specific form [39], [40], whereby the separability between the anomaly and the background is significantly improved for high-precision detection. Methods, such as low-rank tensor decomposition-based detector (LTDD) and prior-based tensor approximation (PTA), are competitive for modeling an HSI being processed as a higher order tensor to exploit both spatial and spectral information to improve the detection accuracy [41], [42], [43], [44], [45]. In addition, deep learning is uniquely advantageous in dealing with diverse, multilevel, and high-complexity data [46], [47], [48]. A wide range of algorithms originating from convolutional neural network (CNN) [49], autoencoder (AE) [50], deep belief network (DBN) [51], and generative adversarial network (GAN) are capable of mining data features through extensive training and extracting the information contained therein sufficiently for detection purposes [52]. The successful migration and application in hyperspectral anomaly detection fully demonstrate the strong parsing ability of machine learning theories in processing data with complexity. However, data-driven, as the current mainstream design concept of machine learning-based methods, inevitably encounters difficulties in training data quality, parameter sensitivity, and computational complexity during the implementation process.

The technical challenge is refined for hyperspectral anomaly detection: in the absence of any priori information, exploit the abnormality to separate the anomaly from the background, and achieve highlighting the anomaly while suppressing the background through the detection output. Such a research idea encourages the exploration of data features in HSIs at the level of mathematical mechanisms, thus providing more possibilities for the design of detection schemes through theoretical migrations [53], [54], [55], [56], [57].

In consideration of both the fixed analytical modes of traditional signal processing-based methods and the difficulties encountered in the costly training process of machine learning-based methods, this article combines point-set topology with information entropy theories to analyze mathematical–statistical properties of hyperspectral data in direct response to the detection requirements for HSIs. The HSI data being processed are mapped into parallel topological spaces in a specific form to achieve ordered arrangements, which helps to deepen the mining of data features. Given that information entropy is a measure of the degree of ordering for a system, with each topological space regarded as a system, the information entropy estimation is performed to drill down into the separable features between the anomaly and the background so as to efficiently extract valid information for detection.

Specifically, the proposed information entropy estimation based on point-set topology (IEEPST) constructs parallel topological spaces on a band-by-band basis through a specific form of mapping to ensure that the anomaly and the background are divided into distinguishable point sets. In this way, the HSI dataset in each topological space is rearranged according to the division of all pixels. In point-set topology theories, the term “cardinality” is defined as the number of elements in a specific point set. For such a random event in a single topological space, a probability space can be obtained from calculations using cardinalities of the point sets in it. The information entropies of all parallel topological spaces are estimated accordingly, and the topological spaces with lower uncertainty of the HSI are rearranged, i.e., the topological spaces with lower information entropies are selected as the optimal separable spaces. Based on the point sets where the testing pixel is located, an anomaly score is calculated from its cardinalities for the detection output. In summary, the contributions of the research work in this article can be summarized as follows.

1) With anomaly detection being tackled as a data mining task and solved by point-set topology theories, the mapping of hyperspectral data to parallel topological spaces

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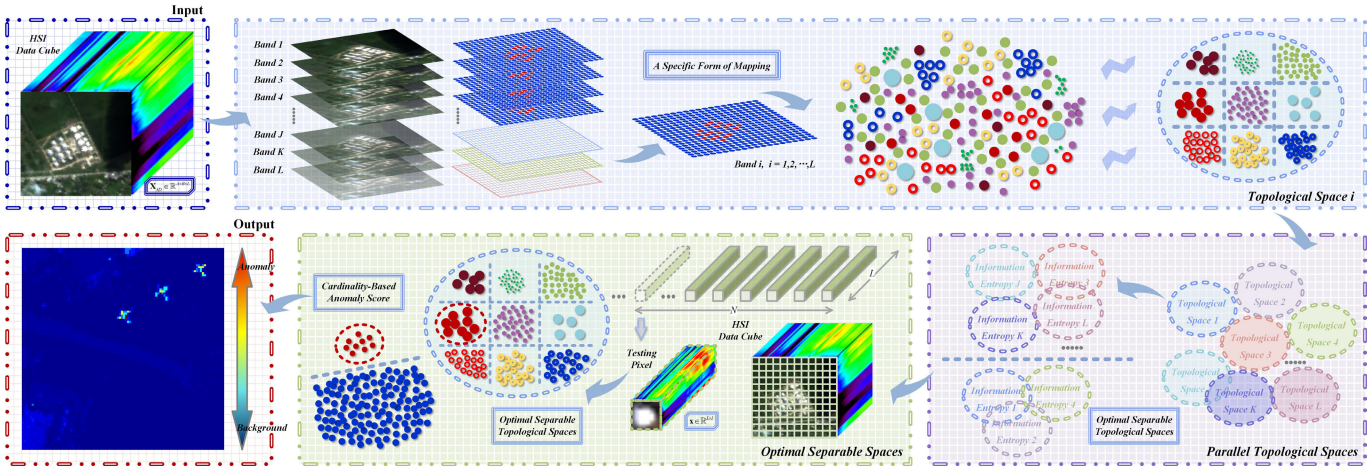


Fig. 1. Flowchart of the proposed IEEPST for hyperspectral anomaly detection.

is a radical departure from the traditional analytical modes to directly mine data features in accordance with detection requirements.

2) The first innovative introduction of information entropy estimation in the adoption of point-set topology fully unravels the ordered arrangement of HSI data in topological spaces. Given that the ordering of data is an entropy-reduction process, the optimal separable spaces are selected to efficiently extract valid land cover information for detection.
3) The selection of optimal separable spaces enables an accurate separation between the anomaly and the background. Moreover, the intuitive manifestation of differences between various classes of land covers in such spaces facilitates the rapid calculation of anomaly scores, which ensures both computational efficiency and superior detection accuracy for the proposed method.

The structure of this article is organized as follows. Section II provides a description of the proposed IEEPST. The experimental results are given in Section III, together with analysis and discussion. The conclusions are drawn in Section IV.

II. Proposed Information Entropy Estimation Based on Point-Set Topology

The processing flow of the IEEPST proposed in this article is shown in Fig. 1, and the specific implementation can be detailed in three parts as follows: 1) the construction of parallel topological spaces using an HSI being processed on a band-by-band basis through a specific form of mapping, where each construction corresponds to an entropy-reduction process; 2) the HSI dataset in each topological space is rearranged according to the division of all pixels, and the optimal separable spaces are selected utilizing the information entropies of the parallel topological spaces; and 3) the anomaly score is calculated for the detection output by tracking the point sets in the optimal separable spaces where the testing pixel is located.

A. Construction of Parallel Topological Spaces

The core idea of applying point-set topology theories to data mining tasks is to map the random, chaotic arrangement of data points in the original space to a topological space where the data points are neatly and orderly arranged due to their data features, facilitating the development of algorithms for specific application requirements [58].

Specifically, for anomaly detection tasks in HSIs without any priori information on availability, pixels (i.e., data points) in the original image (i.e., dataset) are completely random and chaotic in terms of magnitude relationships only, except for the spatial correlation in the imaging scene. An HSI being processed exists in the form of a 3-D data cube, which can be regarded as a collection of pixels at specific locations, each corresponding to a near-continuous spectrum. In the 3-D data cube, each pixel exists as a high-dimensional vector, which is essentially a set of values. The differences in data features reflected in the quantity, density, spectral trend, and morphology of the pixels belonging to various classes of land covers are essentially the differences in the values of various classes of spectra in specific bands. Given that anomaly detection is a binary classification problem, the various classes of land covers present in the HSI can be merely dichotomized into the anomaly and the background, where the anomaly to be detected is of a different class of land cover and occupies a small proportion of the whole image. Therefore, the differences in data features between the anomaly and the background are essentially the differences in the values of the two classes of spectra in specific bands. In other words, the values of anomalous pixels in specific bands are significantly different from those of most surrounding background pixels. Such a “few and different” abnormality of the anomaly provides implementation conditions of pixel-level diagnostics for hyperspectral anomaly detection.

By means of a specific form of mapping, the original HSI data are mapped into the topological space. In the topological space, all pixels are rearranged due to differences in the values in specific bands. As a result, the pixels belonging to various land cover classes differ significantly and intuitively in their

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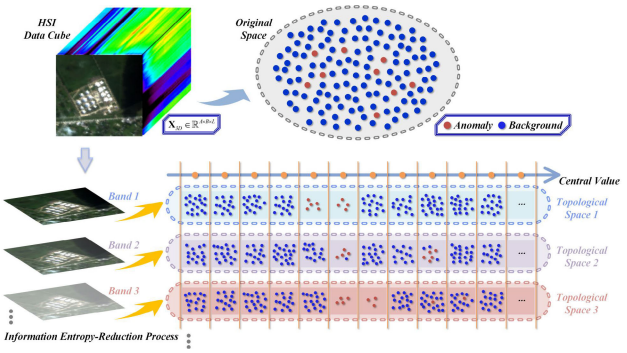


Fig. 2. Illustration of the construction of parallel topological spaces.

distribution due to differences in data features in terms of quantity, density, spectral trend, and morphology, which makes it easier to separate the anomaly from the background. The subsequent design of the detection output to highlight the anomaly while suppressing the background allows for effective detection of the anomaly within a given scene.

As shown in Fig. 2, parallel topological spaces are constructed band-by-band for the HSI being processed, each of which implements a rearrangement of the original dataset. The specific construction is that the minimum–maximum interval of the original HSI is divided equally into multiple parts, each of which corresponds to a central value. With this set of central values, in each band, all pixels are assigned to the corresponding interval according to their closest central value and a set of point sets is generated. Finally, a rearrangement of the complete dataset is achieved once in each band based on numerical differences, thus constructing a topological space where each central value corresponds to a point set. The number of bands in the HSI is equal to the number of parallel topological spaces that are ultimately constructed. In each topological space, the original dataset is ordered once based on the same set of central values to ensure that the rearrangements in all parallel topological spaces are based on the same rule. As mentioned earlier, the ordering of data is an entropy-reduction process and each of the above constructions corresponds to an entropy-reduction process. It is worth noting that the topological space on each band varies in its manifestation and therefore in the information entropy of the rearranged dataset, which can be quantified by the estimation of entropy. In addition, there are cases in Fig. 2 where pixels belonging to the anomaly and background classes are mixed in the same point set. This is a minority of cases due to the fact that no priori information is available during the construction of parallel topological spaces, so it is difficult to guarantee a complete separation between various classes of land covers. In most cases, pixels belonging to various classes of land covers are arranged in point sets separated by relative locations and densities. This provides a solid and reliable basis for the subsequent use of information entropy estimation to select the optimal separable spaces with strong separability between the anomaly and the background.

B. Estimation of Information Entropy

In each topological space, based on a set of central values, all pixels are assigned to respective intervals to achieve an ordered arrangement. Each interval corresponds to a point set, and the sum of the number of pixels in all intervals is equal to the total number of pixels in the HSI being processed. Fig. 3 partially illustrates the ordered arrangement of an HSI in topological spaces corresponding to each band, where the size and color of circles vary according to the cardinalities of point sets. Denote the HSI being processed by $\mathbf{X} \in \mathbb{R}^{L\times N}$, where L is the number of bands and N is the number of pixels. The rearrangement of data in a single topological space is regarded as a collection of information sources $\boldsymbol{Y}_i, i = 1, 2, \ldots, L$, where i denotes the ordinal number of topological spaces, corresponding to the band index in $\mathbf{X}$. The number of point sets in each topological space is denoted by N_{topo}. For example, in each topological space shown in Fig. 4, $N_{\text{topo}} = 9$. The discrete random variable $\boldsymbol{Y}_i$ takes values from $\{y_{i1}, y_{i2}, \ldots, y_{iN_{\text{topo}}}\}$, where y_{ij} represents a random event that a pixel value is assigned to the jth interval in the ith topological space.

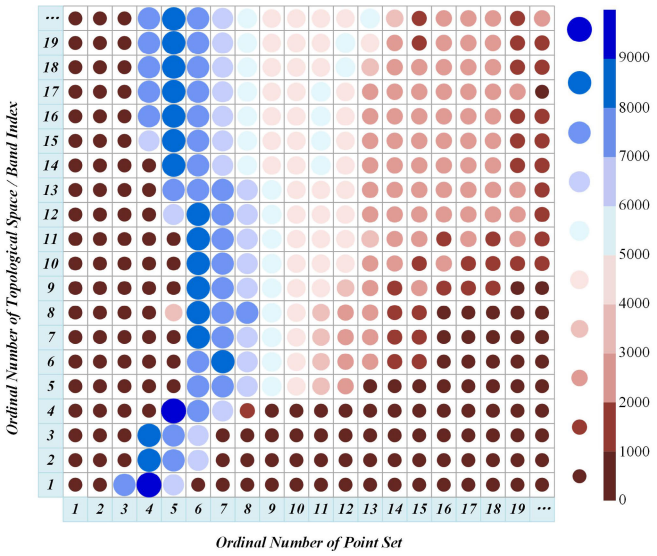


Fig. 3. Ordered arrangement of an HSI in parallel topological spaces.

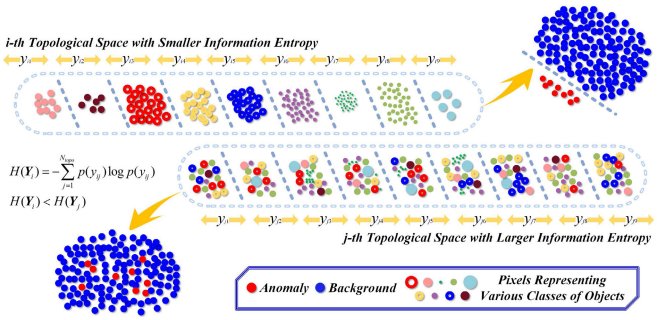


Fig. 4. Topological spaces with varying information entropies.

The formula for calculating information entropy proposed by C. E. Shannon is given as follows:

$$H(\boldsymbol{Y}_i) = E(I(\boldsymbol{Y}_i)) = E[-\log p(\boldsymbol{Y}_i)] = -\sum_{j=1}^{N_{\text{topo}}} p(y_{ij}) \log p(y_{ij}) \quad (1)$$

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where $I(\boldsymbol{Y}_i)$ represents the information content of a random event that may occur in $\boldsymbol{Y}_i$, which is a measure of the uncertainty contained in the occurrence of the event. Also, $p(y_{ij})$ indicates the occurrence probability of y_{ij}, which can be calculated from the cardinality of the corresponding point set. The information entropy $H(\boldsymbol{Y}_i)$ represents the average uncertainty of the occurrence of events in $\boldsymbol{Y}_i$. With regarding the ith topological space as a system, the information entropy of data rearrangement in it measures the degree of ordering of this system. The smaller the information entropy is, the lower the uncertainty of the division of all pixels in the HSI. Accordingly, the smaller the uncertainty of data rearrangement, the more ordered and stable the system corresponding to the topological space in which it is located.

Let $\text{Card}_{ij}, i = 1, 2, \ldots, L, j = 1, 2, \ldots, N_{\text{topo}}$ denote the cardinality of the jth point set in the ith topological space. This leads to a probability space $\boldsymbol{P}_i, i = 1, 2, \ldots, L$, for data rearrangement of this set of random events $\boldsymbol{Y}_i$

$$\begin{bmatrix} \boldsymbol{Y}_i \\ \boldsymbol{P}_i \end{bmatrix} = \begin{bmatrix} y_{i1} & y_{i2} & \cdots & y_{iN_{\text{mpon}}} \\ p(y_{i1}) & p(y_{i2}) & \cdots & p\left(y_{iN_{xpo}}\right) \end{bmatrix} \tag{2}$$

where

$$p(y_{ij}) = \frac{\text{Card}_{ij}}{N}. \tag{3}$$

On this basis, the information entropy of $\boldsymbol{Y}_i$ can be calculated by (1), where the base of the logarithm is set to 2 and the unit of the information content is bit.

With the quantification of the information entropy of data rearrangement in parallel topological spaces, it is feasible to select topological spaces that are favorable for the detection task. It is known that the smaller the information entropy is, the lower the uncertainty, and the higher the degree of ordering, the more stable the data arrangement is. In such spaces, pixels belonging to various classes of land cover differ significantly in their arrangement due to data features, which facilitates the accurate separation of the anomaly from the background, as shown in Fig. 4. Thus, among multiple parallel topological spaces, spaces with smaller information entropies are selected as the optimal separable spaces by specific criteria. Specifically, for the L parallel topological spaces constructed for the HSI $\mathbf{X}$ being processed, $\mathbf{H} = [H(\boldsymbol{Y}_1), H(\boldsymbol{Y}_2), \ldots, H(\boldsymbol{Y}_L)]^{\text{T}} \in \mathbb{R}^{L\times 1}$ records the estimated information entropies. Taking the minimum element (denoted as $H_{\min}$) and the maximum element (denoted as $H_{\max}$) of the vector $\mathbf{H}$, with per_{ie} denoting a set percentage, the threshold used to constrain the information entropy is

$$\eta_{\text{ie}} = H_{\min} + \text{per}_{\text{ie}} \times (H_{\max} - H_{\min}). \tag{4}$$

Besides, when the information entropy of a certain topological space is equal to 0, it indicates that the occurrence probability of one event in the corresponding random-event set is equal to 1, and the occurrence probability of the rest $N_{\text{topo}} - 1$ events is all equal to 0. In other words, in such a topological space, all pixels are divided into the same point set, which is extremely unfavorable to the anomaly-background separation. Given the nonnegativity of the information entropy, the selection criteria for the optimal separable spaces are ultimately set as follows: the information entropy is greater than 0 and less than η_{ie}. Let $\boldsymbol{OSS}$ denote the set of ordinal numbers of the optimal separable space, i.e.,

$$\boldsymbol{OSS} = \{i \in \mathbb{N}^* | 0 < H(\boldsymbol{Y}_i) < \eta_{\text{ie}}\}. \tag{5}$$

Algorithm 1 describes the specific implementation steps in detail.

C. Calculation of Anomaly Scores in Optimal Separable Spaces

At the mathematical statistics level, the most obvious and concrete manifestation of the abnormality lying in the anomaly is "few and different," which provides implementation conditions of pixel-level diagnostics for anomaly detection. Without any priori information, pixels that differ from the majority in the dimensions of quantity, spatial distribution, and spectral characteristics belong to the anomaly class. In pixel-level diagnostics, for a given testing pixel, if it is different in these dimensions, then the pixel can be assumed to be anomalous, and its likelihood of being judged as the anomaly is higher.

In the proposed method, the abnormality of pixels in the HSI can be measured by the designed anomaly scores. As mentioned earlier, there are significant differences in the arrangement of anomaly and background in the optimal separable spaces. In other words, pixels belonging to these two classes differ in their relative locations and densities and are within clearly distinguishable point sets. Therefore, the design for the anomaly scores mainly factors in relative locations and densities in the optimal separable spaces. On this basis, quantization is realized by combining the properties of point sets.

Given that in all topological spaces, the point sets are arranged according to increasing central values, the anomaly and the background are located in separate point sets in terms of relative locations. Furthermore, the background pixels are densely distributed and belong to high-quantity and high-density populations, whereas the anomaly pixels are sparsely distributed and belong to the low-quantity and low-density ones. Thus, in terms of relative densities, the cardinalities of the separate point sets for anomaly and background vary widely. For pixels in a point set, the larger the cardinality, the less anomalous they are, and the more likely being judged as the background. Conversely, the smaller the cardinality of the point set, the more likely pixels in it being judged as the anomaly. On this basis, the computation of anomaly scores in the optimal separable spaces is performed based on cardinalities of point sets.

Assuming that $\mathbf{x} \in \mathbb{R}^{L\times 1}$ is the testing pixel in the HSI $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N] \in \mathbb{R}^{L\times N}$, the distribution of $\mathbf{x}$ in the optimal separable spaces is tracked by the ordinal numbers recorded in $\boldsymbol{OSS} = [\text{OSS}_1, \ldots, \text{OSS}_{N_{\text{oss}}}]^T \in \mathbb{R}^{N_{\text{oss}}\times 1}$, where N_{oss} denotes the number of optimal separable spaces. In each space, the point set in which $\mathbf{x}$ lies is found and its location is recorded in $\mathbf{Coor}_x = [\text{Coor}_{x1}, \ldots, \text{Coor}_{xN_{\text{oss}}}]^{\text{T}} \in \mathbb{R}^{N_{\text{oss}}\times 1}$. With $\mathbf{Card}$, the cardinalities of the point sets in which $\mathbf{x}$ lies in the optimal separable spaces are obtained and recorded in $\mathbf{Card}_x = [\text{Card}_{x1}, \ldots, \text{Card}_{xN_{\text{oss}}}]^{\text{T}} \in \mathbb{R}^{N_{\text{oss}}\times 1}$, where $\text{Card}_{xj} = \mathbf{Card}(\text{OSS}_j, \text{Coor}_{xj})$. It is known that the maxima of the

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Algorithm 1 IEE-OSS ($\mathbf{X}$, N_{topo}, per_{ie})

Input: $\mathbf{X} = [\mathbf{X}_1; \ldots; \mathbf{X}_L] \in \mathbb{R}^{L \times N}$, where $\mathbf{X}_i \in \mathbb{R}^{1 \times N}$-the HSI being processed, N_{topo}-the number of point sets in each topological space, per_{ie}-the percentage used to set the threshold to constrain the information entropy.
Output: $\mathbf{Card} \in \mathbb{R}^{L \times N_{topo}}$-the cardinalities of point sets in parallel topological spaces, $\mathbf{Card}_{max} \in \mathbb{R}^{L \times 1}$-the maxima of the cardinalities of point sets in parallel topological spaces, ***OSS***-the set of ordinal numbers of the optimal separable spaces.
Initialization: $\mathbf{H} = [H_1, \ldots, H_L]^{\mathrm{T}} \in \mathbb{R}^{L \times 1}$, $\mathbf{Card} = [\mathbf{Card}_1; \ldots; \mathbf{Card}_L] \in \mathbb{R}^{L \times N_{topo}}$, where $\mathbf{Card}_i = [Card_{i1}, \ldots, Card_{iN_{topo}}] \in \mathbb{R}^{1 \times N_{topo}}$, $\mathbf{Card}_{max} = [Card_{a1}, \ldots, Card_{aL}]^{\mathrm{T}} \in \mathbb{R}^{L \times 1}$, $\boldsymbol{OSS} = \emptyset$.
1: The minimum-maximum interval of $\mathbf{X}$ is divided equally into N_{topo} parts, each of which corresponds to a central value, and this set of central values is recorded in $\mathbf{Cen} \in \mathbb{R}^{1 \times N_{topo}}$;
2: Start a traversal to construct parallel topological spaces and to estimate the information entropy for the data rearrangement in them;
for $i = 1 : L$ **do**
  3: Denote the set of random events that is the data rearrangement in the current topological space by $\boldsymbol{Y}_i$, $H(\boldsymbol{Y}_i) = 0$;
  4: Given $\mathbf{X}_i$ and $\mathbf{Cen}$, all pixels in the current band are assigned to the corresponding intervals according to their closest central values and N_{topo} point sets are generated;
  5: The cardinalities of N_{topo} point sets are recorded in $\mathbf{Card}_i$;
  6: Find the maximum of the cardinalities of N_{topo} point sets in the current topological space and record it in $Card_{ai}$;
  7: Start a traversal to estimate the information entropy of data rearrangement in the current topological space;
  for $j = 1 : N_{topo}$ **do**
    8: Get the cardinality of the j-th point set in the current topological space, $Card_{ij} = \mathbf{Card}_i(j)$;
    9: The random event that a pixel in the current topological space is divided into the jth point set is represented by y_{ij}. Calculate the occurrence probability of y_{ij} and its information content, $p(y_{ij}) = \frac{Card_{ij}}{N}$, $I(y_{ij}) = -\log p(y_{ij})$;
    10: $H(\boldsymbol{Y}_i) = H(\boldsymbol{Y}_i) + p(y_{ij}) \times I(y_{ij})$;
  11: **end for (step 7)**
  12: $H_i = H(\boldsymbol{Y}_i)$;
13: **end for (step 2)**
14: Take the minimum and maximum of $\mathbf{H}$, H_{min} and H_{max}, and set the threshold using per_{ie}, $\eta_{ie} = H_{min} + per_{ie} \times (H_{max} - H_{min})$;
15: Find ordinal numbers of the optimal separable spaces by means of the selection criteria, $\boldsymbol{OSS} = \{i \,|\, 0 < H(\boldsymbol{Y}_i) < \eta_{ie}\}$;
16: **return** $\mathbf{Card}$, $\mathbf{Card}_{max}$, ***OSS***

cardinalities of point sets in parallel topological spaces are recorded in $\mathbf{Card}_{\max}$. By means of OSS, the maxima of the cardinalities in the optimal separable spaces are obtained and recorded in $\mathbf{Card}_{\text{max-oss}} = [\text{Card}_{m1}, \ldots, \text{Card}_{mN_{\text{oss}}}]^T \in \mathbb{R}^{N_{\text{oss}} \times 1}$, where $\text{Card}_{mj} = \mathbf{Card}_{\max}(\text{OSS}_j)$. Given $\mathbf{Card}_{\text{max-oss}}$ and $\mathbf{Card}_x$, a set of ratios is calculated to obtain $\mathbf{Card}_{\text{max-x-oss}} = [\text{Card}_{mx1}, \ldots, \text{Card}_{mxN_{\text{oss}}}]^T \in \mathbb{R}^{N_{\text{oss}} \times 1}$, where $\text{Card}_{mxj} = (\text{Card}_{mj}/\text{Card}_{xj})$, from which the deviation of $\mathbf{x}$ relative to high-quantity and high-density populations in the optimal separable spaces is measured.

Ultimately, the calculation of anomaly scores for the detection output can be completed

$$D_{\text{IEEPST}}(\mathbf{x}) = \text{AnomalyScore} = \frac{\|\mathbf{Card}_{\text{max-x-oss}}\|_1}{N_{\text{oss}}}. \quad (6)$$

For a testing pixel, the larger the anomaly score, the greater the deviation relative to the high-quantity and high-density populations. Hence, given a threshold η, the following judgment criteria are available:

$$D_{\text{IEEPST}}(\mathbf{x}) \begin{cases} > \eta \rightarrow \text{anomaly} \\ < \eta \rightarrow \text{background}. \end{cases} \quad (7)$$

If $D_{\text{IEEPST}}(\mathbf{x}) > \eta$, then the testing pixel $\mathbf{x}$ is more likely to be judged as belonging to the anomaly class. Conversely, if $D_{\text{IEEPST}}(\mathbf{x}) < \eta$, then $\mathbf{x}$ is more likely to be judged as belonging to the background class. The specific steps are described in Algorithm 2.

In combination with Algorithms 1 and 2, it can be observed that there are two free parameters included in the input of the proposed method. Among them, N_{topo} determines the number of point sets in the parallel topological spaces, while per_{ie} determines the threshold used to constrain the information entropy during the selection of optimal separable spaces. The impact of the parameter settings on the detection effectiveness and computational efficiency is analyzed in Section III.

III. Experimental Results And Analysis

Hyperspectral images covering both natural and artificial environments are employed for the experiments in an attempt to provide an integrative analysis of the performance of the proposed method. Section III-A describes benchmark datasets, evaluation metrics, and experimental configurations, including software and hardware. Section III-B demonstrates and analyzes the impact of key parameters on the performance of the proposed method. Moreover, in Section III-C, representative and state-of-the-art methods are selected for comparison with IEEPST to validate the superior performance of the proposed method.

A. Experimental Preparation

1) Datasets: Five publicly available hyperspectral datasets are selected for the experiments. The information is presented in detail in Table I.

1) The Texas Coast dataset was acquired with the airborne visible/infrared imaging spectrometer (AVIRIS) [59]. An area of Texas Coast with 100 × 100 pixels was collected with a spatial resolution of approximately

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Algorithm 2 IEEPST (**X**, **Card**, $\mathbf{Card}_{\max}$, and ***OSS***)

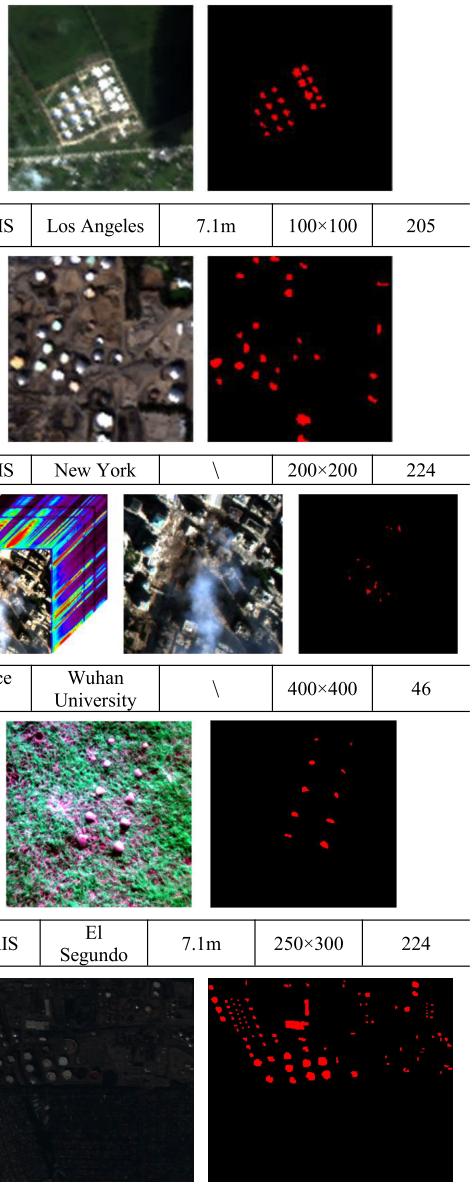
Input: $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N]$, $\mathbf{X} \in \mathbb{R}^{L \times N}$-the HSI being processed, $\mathbf{Card} \in \mathbb{R}^{L \times N_{topo}}$-the cardinalities of point sets in parallel topological spaces,
$\mathbf{Card}_{max} \in \mathbb{R}^{L \times 1}$-the maxima of the cardinalities of point sets in parallel topological spaces,
$\boldsymbol{OSS} = [OSS_1, \ldots, OSS_{N_{oss}}]^T \in \mathbb{R}^{N_{oss} \times 1}$-the set of ordinal numbers of the optimal separable spaces.
Output: $\mathbf{d} = [d_1, \ldots, d_N] \in \mathbb{R}^{1 \times N}$-detection results.
Initialization: $\mathbf{Card}_x = [Card_{x1}, \ldots, Card_{xN_{oss}}]^{\mathrm{T}} \in \mathbb{R}^{N_{oss} \times 1}$.

1: Get the number of elements N_{oss} in ***OSS***, i.e., the number of optimal separable spaces;
2: Obtain the maxima of the cardinalities in the optimal separable spaces through $\mathbf{Card}_{max}$ and ***OSS***, and record them in $\mathbf{Card}_{max\text{-}oss} \in \mathbb{R}^{N_{oss} \times 1}$;
3: Start a traversal to calculate anomaly scores;

for $i = 1 : N$ **do**
4: $\mathbf{x}_i \in \mathbb{R}^{L \times 1} \leftarrow i$-th testing pixel of **X**;
5: Initialize the record of locations of the current testing pixel in the optimal separable spaces, $\mathbf{Coor}_x = [Coor_{x1}, \ldots, Coor_{xN_{oss}}]^{\mathrm{T}} \in \mathbb{R}^{N_{oss} \times 1}$;
6: Start a traversal to track the distribution of the current testing pixel $\mathbf{x}_i$ in the optimal separable spaces;
for $j = 1 : N_{oss}$ **do**
7: Extract the ordinal number of the current optimal separable space $\boldsymbol{OSS}_j$;
8: Get the value of the testing pixel in the current optimal separable space, $x_{ij} = \mathbf{x}_i(OSS_j)$;
9: Find the point set in which x_{ij} is located and record its location in $Coor_{xj}$;
10: Through **Card**, obtain the cardinality of the point set in which $\mathbf{x}_i$ is located in the current optimal separable space, $Card_{xj} = \mathbf{Card}(OSS_j, Coor_{xj})$;
11: **end for (step 6)**
12: By means of $\mathbf{Card}_x$ and $\mathbf{Card}_{max\text{-}oss}$, a set of ratios is calculated to obtain $\mathbf{Card}_{max\text{-}x\text{-}oss} \in \mathbb{R}^{N_{oss} \times 1}$;
13: Average N_{oss} ratios recorded in $\mathbf{Card}_{max\text{-}x\text{-}oss}$ to calculate the anomaly score for the detection output, $d_i = AnomalyScore = \frac{\|\mathbf{Card}_{max\text{-}x\text{-}oss}\|_1}{N_{oss}}$;
14: **end for (step 3)**
15: **return d**

TABLE I
DETAILED INFORMATION OF EXPERIMENTAL DATASETS

	Spectrometer	Capture Location	Spatial Resolution	Spatial Size	Number of Bands
Texas Coast	AVIRIS	Texas Coast	17.2m	100×100	207
Los Angeles	AVIRIS	Los Angeles	7.1m	100×100	205
AVIRIS-WTC	AVIRIS	New York	\	200×200	224
Cri	Nuance Cri	Wuhan University	\	400×400	46
Segundo	AVIRIS	El Segundo	7.1m	250×300	224



AVIRIS: Airborne Visible/Infrared Imaging Spectrometer

17.2 m. After removing the absorption and low signal-to-noise ratio (SNR) bands, 207 spectral channels were retained.

2) The Los Angeles dataset was acquired by the AVIRIS imaging an area of Los Angeles, USA, at a spatial resolution of approximately 7.1 m [60]. There are 100 × 100 pixels in the scene, and 205 bands with better quality were retained for experiments.

3) The AVIRIS-WTC dataset was acquired by NASA's Jet Propulsion Laboratory through the AVIRIS [61]. The imaging location is the World Trade Center (WTC) in New York, with a spatial size of 200 × 200 pixels. Also, there are 224 bands in the wavelength range spanning 370–2510 nm reserved for experiments.

4) The Cri dataset was acquired by imaging an area of Wuhan University with the Nuance Cri hyperspectral sensor [62]. The full image contains 400 × 400 pixels with 46 spectral channels retained at a spectral resolution of 10 nm.

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5) The Segundo dataset was acquired through imaging of an area of El Segundo in California, USA, using AVIRIS [38]. The scene encompasses a park, a petroleum refinery, residential areas, and a school district, with a total of 250 × 300 pixels and a spatial resolution of 7.1 m. In addition, this dataset consists of 224 spectral channels spanning from 366 to 2496 nm. Within this scene, structures, such as storage tanks and towers in the petroleum refinery, are treated as anomalies to be detected.

2) Evaluation Metrics: The experimental results are evaluated using authoritative and recognized metrics in the field, thereby enabling a scientific qualitative and quantitative analysis.

1) Given a set of thresholds, the receiver operating characteristic (ROC) curve visualizes the relationship between probabilities of detection (PDs) and false alarm rates (FARs) [63]. For a detection method, superior performance implies higher PDs and lower FARs. Hence, the more the ROC curve is skewed upward to the left, the better the detection effect.
2) The area under the curve (AUC) is a quantitative metric based on the ROC curve [64]. By integrating the ROC, the AUC is capable of accurately distinguishing subtle differences in detection performance between various algorithms. The larger the AUC, the better the detection performance.
3) In conjunction with the ground truth, the separability map visualizes the separation ability of detection algorithms by demonstrating the statistical range of two sets of values at anomaly and background locations in detection results [65]. With varying colored boxes representing the two classes of land covers, the lower intersection degree of the two colored boxes reflects the stronger capability of separating the anomaly from the background.

3) Experimental Configurations: Regarding the experimental configurations, the software platform is MATLAB R2014a. The hardware device is a Windows 11 computer with a 12th Gen[1] Core[2] i7-1260 CPU @ 2.10 GHz and 16.0-GB RAM.

B. Discussion on Parameters

As mentioned earlier, the algorithmic input to the proposed IEEPST contains two key parameters: the number of point sets N_{topo} in each topological space and the percentage per_{ie} used to calculate the threshold to constrain the information entropy. From the data information provided in Table I, the HSIs used in experiments vary in spatial resolution and size. Therefore, choosing reasonable ranges for the above two parameters is not only for searching the optimal combinations but also for observing the impact of varying settings on detection effects through extensive experimental results. Evaluation of the detection results using AUC, a quantitative metric, facilitates an integrated, comprehensive, and efficient analysis of the parameter sensitivity for the proposed method. For N_{topo}, the setting range is [10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200]. It is worth noting that there is no specific constraint between the setting of N_{topo} and the number of pixels in an HSI being processed. In the construction of a single topological space, all pixels in a specific band are assigned to the corresponding interval according to their closest central value and N_{topo} point sets are generated. Therefore, the setting of N_{topo} is not limited by the number of pixels in the HSI. As for per_{ie}, the range is set to [10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%]. Ultimately, there are a total of 200 parameter combinations on each dataset, and thus, 200 sets of detection results are to be recorded and evaluated. Such a wide range of parameter settings ensures that the contingency caused by the parameter discussion with a few experimental results is avoided, which is favorable for deriving more reliable and practical conclusions.

1) Detection Effect: For AUCs obtained under varying parameter combinations, the magnitude of values is used to color bar charts to illustrate how the detection effect varies with parameter settings. The experimental results on Texas Coast, Los Angeles, AVIRIS-WTC, Cri, and Segundo are plotted in Fig. 5(a)–(e), respectively. Besides, based on peak points of AUCs, the optimal parameter combinations for IEEPST on all experimental datasets are given in Table II for reference. Regarding the parameter sensitivity of the proposed method, the following conclusions can be drawn.

TABLE II
RECOMMENDED PARAMETER SETTINGS FOR IEEPST ON EXPERIMENTAL DATASETS

	Evaluation Result (AUC)	Recommended Parameter Setting	
		N_{topo}	per_{ie}
Texas Coast	0.9989	20	90%
Los Angeles	0.9940	30	100%
AVIRIS-WTC	0.9980	140	40%
Cri	0.9998	20	60%
Segundo	0.9903	140	60%

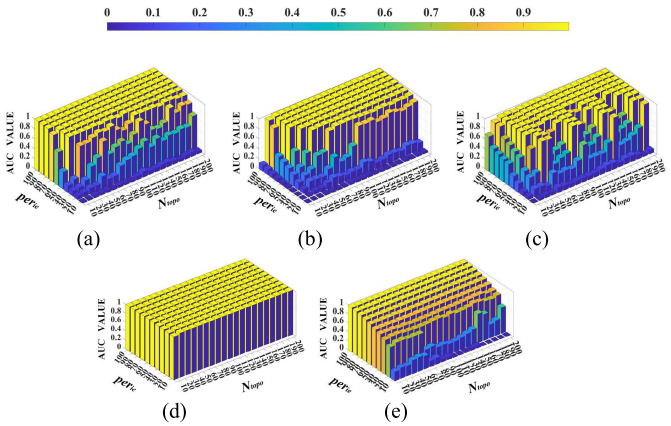


Fig. 5. Colored bar charts illustrating the relationship between combinations of N_{topo} and per_{ie} and the detection effect of IEEPST on experimental datasets. (a) Texas Coast. (b) Los Angeles. (c) AVIRIS-WTC (d) Cri. (e) Segundo.

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1) As N_{topo} and per_{ie} increase, the AUC values gradually increase and the variation decreases and finally converge. It can be observed that the performance of the proposed IEEPST is robust under reasonable setting ranges of parameters. Equipped with a low parameter sensitivity, it facilitates IEEPST to perform anomaly detection tasks on HSIs with diversity and to meet the requirements for detection response with high PDs and low FARs.
2) In conjunction with the optimal parameter combinations given in Table II, it is feasible to explore setting rules for various types of imaging scenes. For HSIs with relatively homogeneous distributions of imaging scenes such as Texas Coast, Los Angeles, and Cri, the AUC of the proposed IEEPST peaks when N_{topo} is smaller and per_{ie} is larger. For HSIs with more complex imaging scenes such as AVIRIS-WTC and Segundo, the AUC peaks at a larger N_{topo} and a smaller per_{ie}.

Integrating the experimental results in the parameter discussion part, it can be concluded that the proposed IEEPST is less sensitive to the parameter settings and exhibits robustness. Moreover, for imaging scenes with different types of distributions in practical applications, recommendations can be provided based on the parameter setting rules summarized earlier. Such rules can be explained along with the mechanism of the proposed method described in Section II. During the construction of parallel topological spaces, pixels belonging to various classes of land covers are divided into different point sets due to their data features. Therefore, for imaging scenes with relatively homogeneous distributions, it is recommended to set N_{topo} smaller to fit fewer classes of land covers. Furthermore, a homogeneous distribution of land covers implies less interference with the separation of the anomaly and the background, and more topological spaces are available to achieve an accurate separation of the two classes of land covers. Hence, it is recommended to set per_{ie} larger to loosen the information entropy constraints to retain more optimal separable spaces. Conversely, for imaging scenes with more complex distributions, a larger N_{topo} should be set to fit the wide variety of land covers. In this case, too much topological spaces may interfere with separating the anomaly from the background, so it is necessary to tightly constrain the information entropy to reduce the number of optimal separable spaces by setting per_{ie} smaller.

2) Computational Efficiency: Along with the given range of parameter settings, the execution time of the proposed method is recorded and surfaces are colored with the magnitude of values to demonstrate the computational efficiency, as shown in Fig. 6. In particular, the complete steps corresponding to Algorithms 1 and 2 in the proposed method are incorporated into timing. It can be observed that the surfaces on all experimental datasets are near-sloped, albeit with slight fluctuations due to equipment-induced systematic errors. The execution time of IEEPST increases roughly linearly as N_{topo} and per_{ie} increase, indicating that the proposed method is equipped with a near-linear time complexity. In addition, compared to N_{topo}, per_{ie} exerts a more significant effect on computational efficiency, which is conducive to the efficient completion of the parameter tuning process. Without any training samples with labels and a costly and tedious training process, the proposed IEEPST is favorable to be extended to practical applications to respond quickly to various detection requirements and demonstrate its unique advantages.

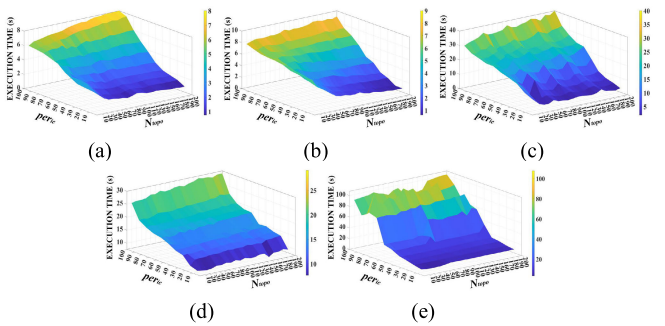


Fig. 6. Colored surfaces illustrate the relationship between combinations of N_{topo} and per_{ie} and the computational efficiency of IEEPST on experimental datasets. (a) Texas Coast. (b) Los Angeles. (c) AVIRIS-WTC. (d) Cri. (e) Segundo.

C. Comparison of Performance

After discussing the key parameters and summarizing the setting rules applicable to different imaging scenes, the proposed method is compared with other representative and state-of-the-art detection methods to verify its effectiveness and superiority. The methods selected to participate in the comparative experiments are RXD (Global-RXD) [18], Local-RXD [63], robust principal component analysis RXD (RPCA-RXD) [66], CRD [27], LRASR [28], graph total variation regularized low-rank representation-based detector (GTVLRR) [67], and two-step GLRT (2S-GLRT) [20]. Considering that some of the above algorithms adopt the concentric dual-window detection mode, the sizes of inner and outer windows are properly set based on the information of the experimental datasets to ensure that the methods involved in the comparison can achieve competitive detection effects. On Texas Coast and AVIRIS-WTC, the sizes of the outer and inner windows for both are set to 15×15 and 5×5, respectively. For Los Angeles, the sizes of the two windows are 17×17 and 7×7. For Cri, the outer window is set to 25×25 and the inner window is set to 15×15. For Segundo, the sizes of two windows are 21×21 and 11×11. Comparisons on all HSI datasets are performed on the same hardware and software platforms, and the experimental results are evaluated with the same metrics to ensure fairness.

1) Detection Effect: In Fig. 7, the detection results of all the methods involved in the comparison are linearly stretched to the range of 0–255, and detection maps are plotted by coloring the values according to their magnitude. By visually comparing the detection results, the following conclusions can be drawn.

1) Land covers belonging to the background class are highlighted to varying degrees in the detection maps of Global-RXD, RPCA-RXD, LRASR, and GTVLRR. The presence of contours of farmland, grass, and artificial buildings indicates that the methods are weak at suppressing the background.

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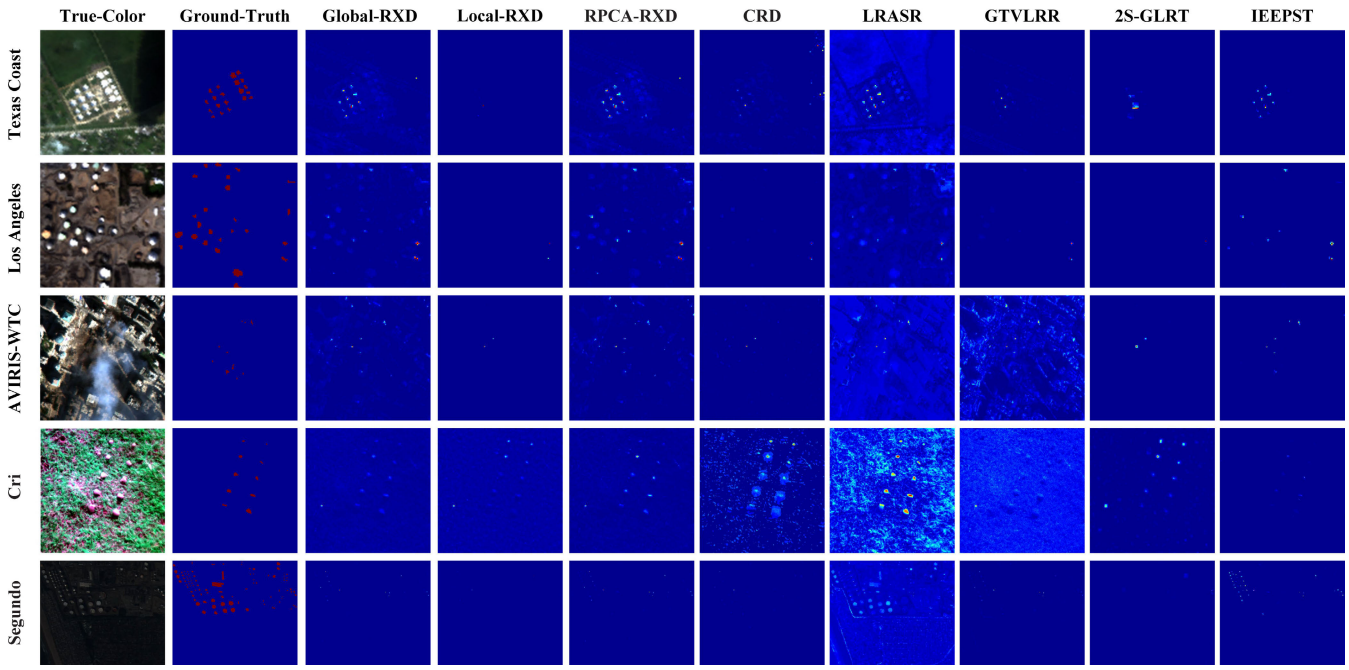


Fig. 7. Colored maps for comparison between detection methods on experimental datasets.

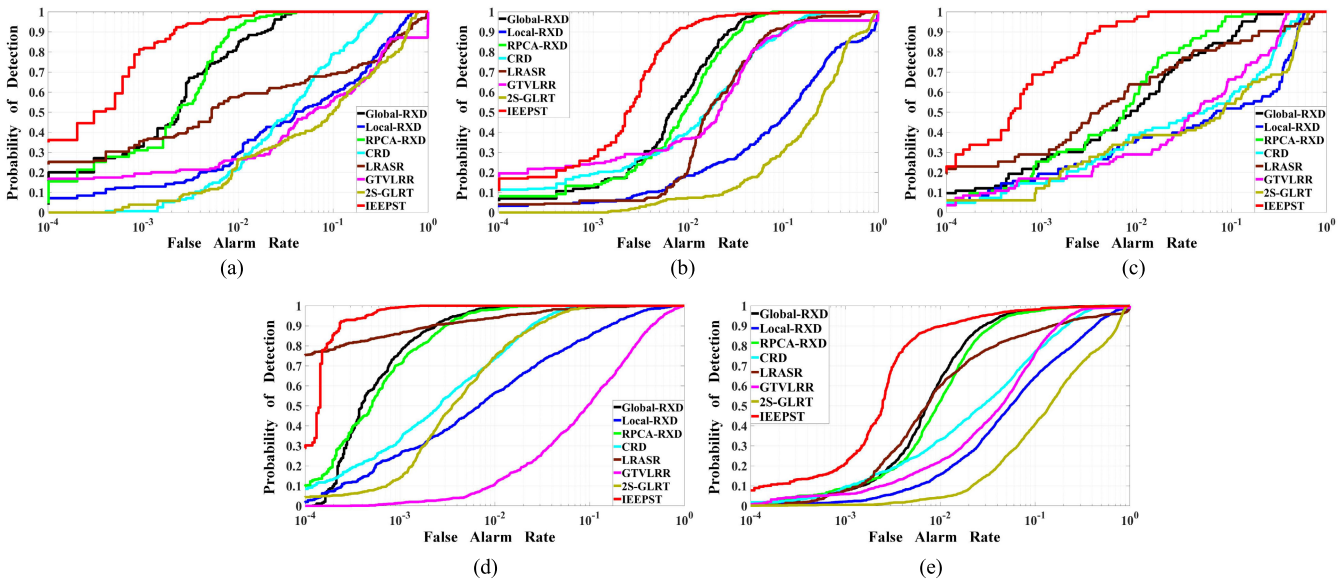


Fig. 8. ROC curves for comparison between detection methods on experimental datasets. (a) Texas Coast. (b) Los Angeles. (c) AVIRIS-WTC. (d) Cri. (e) Segundo.

2) The detection maps of Local-RXD, CRD, and 2 S-GLRT exhibit lower values in the background portion, reflecting a stronger capability of suppressing the background interference. However, these three algorithms are unsatisfactory in highlighting the anomaly. The lower values at anomaly locations in the detection results lead to difficult recognition.
3) Comprehensively considering PD and FAR, IEEPST outperforms other algorithms. It not only suppresses the background interference to a very low level in the imaging scenes but also highlights anomaly-class land covers and largely preserves their shapes and contours, resulting in the most visually impressive results.

With the evaluation of the detection results on Texas Coast, Los Angeles, AVIRIS-WTC, Cri, and Segundo, the ROC curves on the above experimental datasets are plotted in Fig. 8(a)–(e), respectively. Synchronously, the AUC values of the detection results for all algorithms are given in Table III, and the optimal evaluation results on each dataset and their corresponding algorithms are marked in bold. In combination with the quantitative and qualitative evaluations of the experimental results, reasonable analyses are carried out to draw conclusions.

1) The ROC evaluation results for Local-RXD and 2 S-GLRT underperform overall, with the curves closer to the bottom right on the experimental datasets.

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TABLE III
EVALUATION RESULTS FOR COMPARISON BETWEEN DETECTION METHODS ON EXPERIMENTAL DATASETS

	Global-RXD	Local-RXD	RPCA-RXD	CRD	LRASR	GTVLRR	2S-GLRT	**IEEPST**
Texas Coast	0.9945	0.7629	0.9959	0.9328	0.8452	0.7766	0.7639	**0.9989**
Los Angeles	0.9886	0.5914	0.9864	0.9645	0.9546	0.9295	0.5313	**0.9940**
AVIRIS-WTC	0.9670	0.7180	0.9813	0.8677	0.9248	0.8939	0.7516	**0.9980**
Cri	0.9990	0.9493	0.9987	0.9916	0.9956	0.8269	0.9900	**0.9998**
Segundo	0.9841	0.8570	0.9810	0.9290	0.9287	0.9244	0.7791	**0.9903**

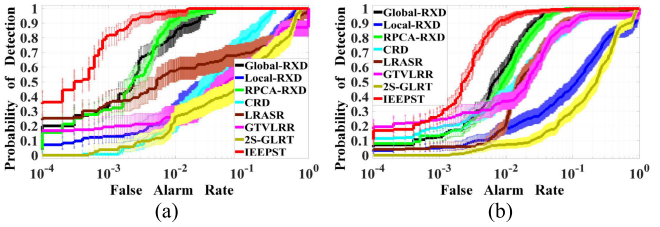


Fig. 9. ROC curves with 95% confidence intervals drawn around each empirical point for continuously varying thresholds. (a) Texas Coast. (b) Los Angeles.

2) The ROC curves of CRD, LRASR, and GTVLRR are intersecting and mediocre, suggesting that these algorithms exhibit varying degrees of disadvantage on some of the datasets.
3) In contrast, the ROC evaluations of Global-RXD, RPCA-RXD, and IEEPST are more prominent, with curves closer to the upper left, demonstrating superior detection effects. In particular, the curves of IEEPST are closest to the upper left on all experimental datasets, thus leading to the overall performance of the other algorithms.
4) Integrating the AUC evaluation results in Table III for quantitative analysis, it can be found that Global-RXD and RPCA-RXD with large AUC values perform second only to IEEPST, which corresponds to the performance of the ROC curves in Fig. 8. Also, IEEPST is evaluated as the best on all experimental datasets, ahead of other algorithms by a huge margin.

In particular, considering that the imaging scenes corresponding to Texas Coast and Los Angeles are relatively small in all experimental datasets, in order to ensure a reliable FAR estimation to drill down to analyze and evaluate the performance of the proposed method at a statistical level, ROC curves with confidence intervals are estimated, whereby intuitive comparisons are made between the predicted and empirical performance [68]. Fig. 9 illustrates the ROC curves with 95% confidence intervals on Texas Coast and Los Angeles datasets with relatively small imaging scenes, and the confidence intervals are estimated with continuously varying thresholds. It can be observed that in addition to the leading PD in the low FAR region, the bounds of the confidence intervals of IEEPST on the ROC curve are also significantly higher than those of the other algorithms involved in the comparison. The estimation of confidence intervals based on ROC curves further demonstrates that the superiority and robustness of the proposed method are statistically significant.

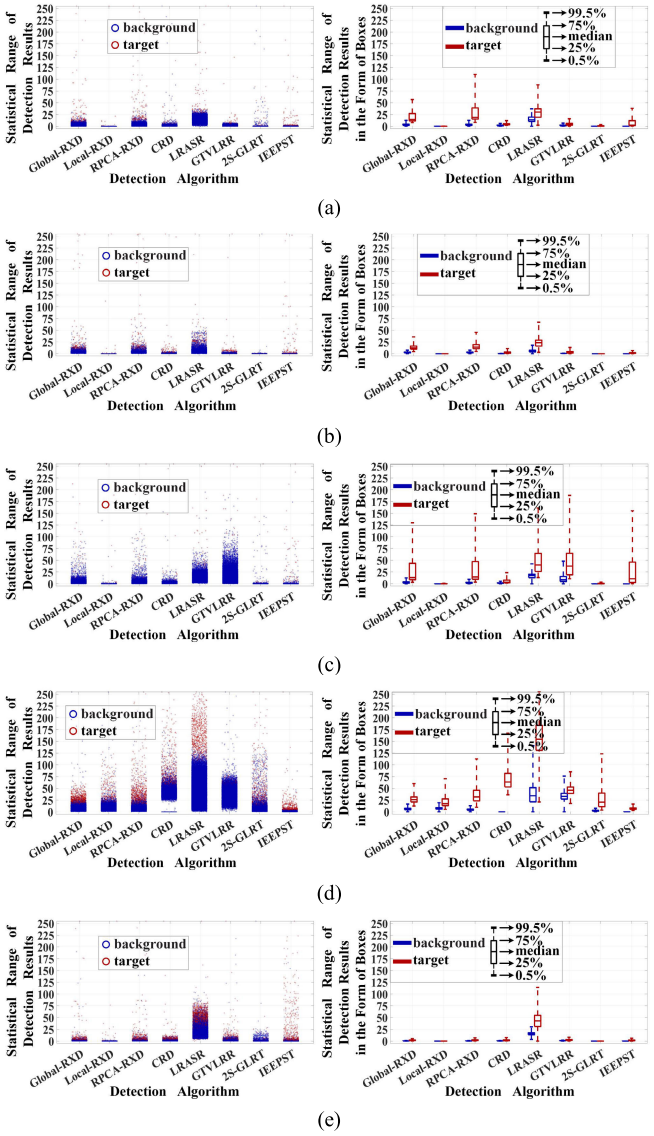


Fig. 10. Separability map for comparison between detection methods on experimental datasets. (a) Texas Coast. (b) Los Angeles. (c) AVIRIS-WTC. (d) Cri. (e) Segundo.

Fig. 10(a)–(e) visualizes the ability of all detection methods to separate the anomaly from the background on Texas Coast, Los Angeles, AVIRIS-WTC, Cri, and Segundo. Based on the location information of the anomaly and the background provided by the ground truth for each dataset, the subfigures on the left illustrate statistical ranges of the two sets of values in detection results, and the subfigures on the right illustrate

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TABLE IV
EXECUTION TIME FOR COMPARISON BETWEEN DETECTION METHODS ON EXPERIMENTAL DATASETS

	Global-RXD	Local-RXD	RPCA-RXD	CRD	LRASR	GTVLRR	2S-GLRT	IEEPST
Texas Coast	0.0681	41.6154	4.8863	83.2156	5.1702	244.4403	57.5185	5.3060
Los Angeles	0.0716	51.4563	4.1320	79.6270	5.3884	213.9515	58.6127	4.2210
AVIRIS-WTC	0.1743	223.6990	24.1475	383.1790	119.1328	398.4943	243.7434	19.0338
Cri	0.0973	138.6563	8.3589	2994.3433	155.9925	2167.3487	7029.7534	17.4108
Segundo	0.8082	554.0906	51.9390	788.8857	39.8382	516.7862	728.2699	99.8153
Summed Time	1.2195	1009.5176	93.4637	4329.2506	325.5221	3541.021	8117.8979	145.7869

the corresponding box-and-whisker plots. The following conclusions are revealed through observations.

1) The values at background locations in the detection results of Global-RXD, RPCA-RXD, LRASR, and GTVLRR are relatively high. With strong background interferences, improper threshold settings can easily lead to false alarms.
2) For Local-RXD, CRD, and 2 S-GLRT, it can be seen that the values at background locations are suppressed to extremely low levels. However, the values at anomaly locations in the detection results are stretched to a similar range as the background, which leads to an inability to separate the anomaly from the background.
3) In contrast, IEEPST is not only able to suppress values at background locations to a very low level but also able to stretch values at anomaly locations to a higher range. The evaluation results of separability map provide ample evidence that IEEPST is capable of separating these two classes of land covers and is effective in highlighting the anomaly while suppressing the background, which echoes the superior visual effects in Fig. 7.

2) Computational Efficiency: With timing from the same start to cover the complete execution to ensure fairness, Table IV summarizes the execution time of all algorithms to compare the computational efficiency. By summing the execution time on all experimental datasets in the last row, it can be concluded that the computational efficiency of the proposed IEEPST is second only to the traditional signal processing-based Global-RXD and RPCA-RXD, which far exceeds that of machine learning-based methods such as CRD, GTVLRR, and 2 S-GLRT. It is further demonstrated that IEEPST with a near-linear time complexity is inexpensive to run, computationally efficient, and highly competitive in the implementation of pixel-level diagnostics for hyperspectral anomaly detection in real imaging scenes with diversity.

IV. CONCLUSION

Aiming at the difficulties of existing methods in terms of modeling mechanism and computational cost, this article introduces the information entropy estimation in the adoption of point-set topology theories for the first time and proposes an interesting anomaly detection method to deeply analyze the mathematical–statistical properties of hyperspectral data and efficiently extract the land cover information for detection. The proposed IEEPST starts by constructing parallel topological spaces to achieve ordered arrangements of HSI data to facilitate the separation of pixels belonging to various land covers. Afterward, information entropies of the parallel topological spaces are estimated to adequately unravel the data rearrangements therein to select the optimal separable spaces between the anomaly and the background. Ultimately, the properties of point sets in the optimal separable spaces are incorporated to complete the calculation of anomaly scores, from which the abnormality of the testing pixel is measured for the detection output. Hyperspectral images with diverse imaging scenes are selected for experiments to fulfill the parameter discussion and the comparison of detection performance. The evaluation results demonstrate that the proposed IEEPST is ahead of other comparative algorithms in terms of comprehensive competitiveness involving parameter sensitivity, computational efficiency, and detection effect. The superior performance of the proposed IEEPST makes it user-friendly for fast response to high-precision detection requirements in real scenes, which is conducive to its further popularization and application.

In the following work, we will try more classic topological structures such as binary tree and attempt to introduce the ensemble learning theory to gradually reduce or even remove the influence of the parameter tuning process on the detection effect. In addition to the research on improving the robustness of the proposed method, experiments will be conducted on more hyperspectral data with diversity.

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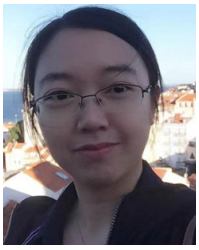
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